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DIU
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Welcome

To

Dhaka International University

DEPARTMENT OF CSE

Instructor	: ANTU CHOWDHURY
Batch	: 93 (1st Semester)
Course Code	: 0613-101
Course Title	: Structured Programming Languages

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Slide Title:

- **Title: Matrix Addition & Subtraction in C Programming**

Introduction

Definition of 2D Matrices:

- A 2D matrix is a collection of numbers arranged in rows and columns.

Rows and Columns:

- Rows: Horizontal arrangement of elements in a matrix.
- Columns: Vertical arrangement of elements in a matrix.

Example

$$\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix}$$

2d matrix

```
int matrix[3][3] =
{
    {1, 2, 3},
    {4, 5, 6},
    {7, 8, 9}

};
```

2d array

Problem

Analysis

Problem name

Matrix Addition & Subtraction

To Do

- We need two 2D array as matrix.
- Subtract corresponding elements of two matrices.
- Add corresponding elements of two matrices.
- Output the Addition & Subtraction result.

ALGORITHM

Input:
Prompt the user to enter the number of rows and columns for matrices A and B.

Input Elements for Matrix A:
Use nested loops to accept elements for matrix A from the user.

Input Elements for Matrix B:
Use nested loops to accept elements for matrix B from the user.

Print Matrix A:
Display the elements of matrix A.

Print Matrix B:
Display the elements of matrix B.

Matrix Addition:
Use nested loops to add corresponding elements of matrices A and B, storing the result in matrix C.

Print Matrix C (Result of Addition):
Display the elements of matrix C, which represents the result of matrix addition.

Matrix Subtraction:
Use nested loops to subtract corresponding elements of matrix B from A, storing the result in matrix C.

Print Matrix C (Result of Subtraction):
Display the elements of matrix C, which represents the result of matrix subtraction.

End:
End the program.

Code Structure:

```c
#include<stdio.h>
int main ()
{
    int i,j,noofrows,noofcols;
    int A[10][10],B[10][10],C[10][10] ;
    printf("Enter the no of rows and cols:");
    scanf("%d %d",&noofrows,&noofcols);
    printf("Enter elements for be A matrix.\n");
    for(i=0; i<noofrows; i++)
        //scanner
    {
        for (j=0; j<noofcols; j++)
        {
            printf("A[%d][%d]:",i,j);
            scanf("%d",&A[i][j]);
        }
        printf("\n");
    }
    printf("Enter elements for be B matrix.\n");
    for(i=0; i<noofrows; i++)
        //scanner
    {
        for (j=0; j<noofcols; j++)
        {
            printf("B[%d][%d]:",i,j);
            scanf("%d",&B[i][j]);
        }
        printf("\n");
    }
    //print of A matrix
    printf("A: ");
    for (i=0; i<noofrows; i++)
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",A[i][j]);
        }
        printf("\n");
    }
    //print of B matrix
    printf("B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",B[i][j]);
        }
        printf("\n");
    }
'addition of matrix
    for (i=0; i<noofrows; i++)
    {
        for(j=0; j<noofcols; j++)
        {
            C[i][j] =A[i][j] +B[i][j] ;
        }
    }
    //print in C matrix
    printf("\n");
    printf("A+B:");
    printf("\n");
    printf("A+B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",C[i][j]);
        }
        printf("\n");
    }
    for (i=0; i<noofrows; i++)
    {
        for(j=0; j<noofcols; j++)
        {
            C[i][j] =A[i][j] - B[i][j] ;
        }
    }
    printf("\n");
    printf("A+B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",C[i][j]);
        }
        printf("\n");
    }
}
```

Explaination

Code Walkthrough

```c
int i,j,noofrows,noofcols;
int A[10][10],B[10][10],C[10][10] ;
```

Declaring the variables

```c
printf("Enter elements for be A matrix.\n")
for(i=0; i<noofrows; i++)
    //scanner
{
    for (j=0; j<noofcols; j++)
    {
        printf("A[%d][%d]:",i,j);
        scanf("%d",&A[i][j]);
    }
    printf("\n");
}
```

```
Enter elements for be A matrix.
A[0][0]:1
A[0][1]:2

A[1][0]:3
A[1][1]:4
```

```c
printf("Enter elements for be B matrix.\n
for(i=0; i<noofrows; i++)
    //scanner
{
    for (j=0; j<noofcols; j++)
    {
        printf("B[%d][%d]:",i,j);
        scanf("%d",&B[i][j]);
    }
    printf("\n");
}
```

```
Enter elements for be B matrix.
B[0][0]:5
B[0][1]:6

B[1][0]:7
B[1][1]:8
```

Explaination

Code Walkthrough

```c
//print of A matrix

printf("A:  ");
for (i=0; i<noofrows; i++)

{
    printf("\t");
    for(j=0; j<noofcols; j++)
    {
        printf("%d ",A[i][j]);

    }
    printf("\n");


}
```

```
A:      1 2
        3 4
```

```c
//print of B matrix
printf("B:");
for (i=0; i<noofrows; i++)
{
    printf("\t");
    for(j=0; j<noofcols; j++)
    {
        printf("%d ",B[i][j]);
    }
    printf("\n");
  }
```

```
B:      5 6
        7 8
```

Code Walkthrough

```c
//addition of matrix
    for (i=0; i<noofrows; i++)
    {
        for(j=0; j<noofcols; j++)
        {
            C[i][j] =A[i][j] +B[i][j] ;
        }
    }
    //print in C matrix
    printf("\n");
 printf("A+B:");
    for (i=0; i<noofrows; i++)
    {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
            printf("%d ",C[i][j]);
        }
        printf("\n");
    }
```

```
A+B:    6 8
        10 12
```

```c
    for (i=0; i<noofrows; i++)

     {
        for(j=0; j<noofcols; j++)
        {
           C[i][j] =A[i][j] - B[i][j] ;
        }
     }
     //print in C matrix
     printf("\n");
  printf("A-B:");
     for (i=0; i<noofrows; i++)
     {
        printf("\t");
        for(j=0; j<noofcols; j++)
        {
           printf("%d ",C[i][j]);
        }
        printf("\n");
     }
```

```
A-B:    -4 -4
        -4 -4
```

CODE OUTPUT:

```
Enter elements for be B matrix.
B[0][0]:5
B[0][1]:6

B[1][0]:7
B[1][1]:8

A:      1 2
        3 4
B:      5 6
        7 8

A+B:    6 8
        10 12

A-B:    -4 -4
        -4 -4
```

OUTPUT

Conclusion

- **Summary:**
  - **We successfully demonstrated the addition and subtraction operations on matrices using C programming.**
  - **Understanding basic concepts of matrices and programming structures is crucial for solving such problems effectively.**

THANK YOU ALL!